NICOLAS RUTH

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EDUCATION

University of Denver — Denver, CO *Expected May 2027*

B.S. in Electrical Engineering — Specialization in Robotics · Minor in Mathematics

- **Relevant coursework:** Advanced Electronics, Physical Electronics, Signals & Systems, Microprocessors, Electromagnetics

EXPERIENCE

RF Engineering Intern — Arsenal Nexus, Denver, CO *June – August 2026*

- Simulated antenna cans in MATLAB to optimize directivity, and designed and simulated Vivaldi antennas ahead of physical build.
- Took the designs to hardware rather than stopping at simulation: azimuth pattern tests to confirm directivity, plus S11 and S21 measurements on a Rigol RSA5065N in VNA mode.

Lead Cashier — Miss Molly's Bakery, Denver *(2022–2023)* — sole closing employee; trained new cashiers and cut wait times.

Landscaper — Self-Employed, Denver *(2022–2023)* — maintained grounds and operated equipment, managing client schedules.

PCB DESIGN

3-Wheel Robot Control Board KiCad · STM32F401 · Hand-soldered SMD

- Designed the schematic and layout in KiCad for a robot control board around an STM32F401RBT6 (Cortex-M4): TB6612FNG motor driver, MPU6050 IMU, MT6701 magnetic encoders, ESP8266 radio, USB-C input behind a PTC fuse and USBLC6 ESD array, and LD1117 regulators feeding separate 5 V and 3.3 V rails.
- Hand-soldered every surface-mount part on the bare board — passives, both regulators, motor driver, crystal, and the STM32 in a 64-pin QFP — spacing the layout for hand assembly and keeping load-bearing connectors through-hole.
- Added a switch cutting the motor rail but leaving logic USB-powered, so the board flashes without driving off the bench.

DemoSat High-Altitude Balloon Payload KiCad

- Redesigned an existing balloon payload as an Arduino Uno shield in KiCad after finding two of its four sensors out of production, choosing replacements against the flight's altitude, temperature, and low-pressure limits.
- Integrated temperature (DS18B20, 1-Wire), humidity (HIH-9120), pressure (DPS310), and 3-axis accelerometer (ADXL345) sensors with SD logging on one shared 3.3 V rail and I²C bus — ~120 g, ~$115, 259 mW typical, against budgets of 200 g and $147.

Arduino Uno I/O Shield KiCad · Built & hand-soldered

- Designed a mixed-signal Uno shield giving the Arduino a true analog output: a 10-bit R-2R ladder DAC (10 kΩ / 20 kΩ) on D0–D9 buffered by an MCP6002 rail-to-rail op-amp, with a DC-offset trimmer and 3.5 mm line in / line out jacks.
- Added two 100 kΩ potentiometers, two buttons, and I²C headers for an MPU6050 IMU and a 128 × 64 SSD1306 OLED.
- Had the board fabricated and hand-soldered the assembly.

Battery Tester Board KiCad

- Designed an eight-channel battery load tester that reads a cell under a switched 20 Ω load rather than open-circuit: two LM324 quad op-amps as eight comparators against a tapped 1 kΩ resistor ladder, with a 5 kΩ trimmer calibrating all eight thresholds together.

TECHNICAL PROJECTS

Electromagnetic Rail Launcher (Railgun) Power Electronics · High Voltage

- Built a rail-type electromagnetic launcher: a DC boost converter steps an 11.1 V LiPo to ~225 V, charging a 13,200 μF bank (six 250 V / 2200 μF electrolytics in parallel, ~330 J stored) discharged across dual aluminum rails to accelerate a small metal projectile.
- Designed a 2.6 kΩ inrush-limiting resistor network with a 600 V blocking diode to protect the charging stage.

FM Radio Receiver RF / Analog

- Built an FM receiver from discrete parts: a hand-wound 0.15 μH coil and 1–30 pF variable capacitor tuning the 88–108 MHz band, an S9018 RF transistor recovering the signal, and an LM386 driving a speaker.
- Diagnosed unresponsive tuning — on a solderless breadboard, stray parasitic capacitance rivalled the 1–30 pF tuning range, so layout rather than the variable capacitor set the resonant frequency.

Electromagnetic Pulse (EMP) Demonstrator High Voltage

- Designed a capacitor-discharge EMP generator: a 9 V high-voltage step-up converter charges a 20 kV ceramic capacitor bank, discharged through a hand-wound coil to produce a pulsed magnetic field.

TECHNICAL SKILLS

Languages: C / Embedded C, Python, MATLAB, Java

Tools: KiCad, LTspice, MATLAB, STM32CubeIDE, Arduino IDE, SolidWorks / Onshape, Microsoft Office

Hardware: STM32, Arduino and Teensy MCUs; schematic capture and PCB layout; hand SMD assembly to 64-pin QFP; component selection; oscilloscope, multimeter, spectrum analyzer, vector network analyzer; 3D printers

Concepts: Analog & digital circuit design, PCB design & hand assembly, robotics, RF and antenna design, sensor integration & data logging, power electronics, electromagnetics, embedded systems

LEADERSHIP & ACTIVITIES

Member — DU IEEE Chapter *(2026–Present)* — assisted in designing a PCB layout for a slice of the chapter's LED matrix cube.

Mechanical Team Lead — FIRST Robotics *(2021–2023)* — led the mechanical subteam, training new members on machines, components, and tools to deliver a competition-ready robot.

Also: DU Time Attack Motorsport Team *(2025)* — vehicle aero and downforce discussions; Mile High Flight Club *(2022–2023)* — aviation and aerospace careers; Colorado Young Leaders *(2023)* — community service and environmental sustainability events.